Junghoon Chung

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SUMMARY

PhD candidate in Industrial & Systems Engineering with a strong background in UX research, human factors, and data-driven design. Experienced in designing and executing end-to-end mixed-methods research including interviews, surveys, controlled experiments, and field studies. Skilled at integrating qualitative insights with large-scale behavioral and sensor data to inform product and system design. Proven ability to translate complex findings into actionable recommendations for cross-functional teams. Seeking a UX Research internship to support evidence-driven product development at scale.

EDUCATION

University of Washington, Seattle Sep 2025 – Jun 2029 (Expected)
Ph.D in Industrial and Systems Engineering, Dean's Fellowship Recipient

Virginia Tech, Blacksburg Aug 2023 – May 2025
M.S in Industrial and Systems Engineering (Thesis Track), GPA 3.89/4.0

Yonsei University, Seoul Mar 2017 – Feb 2023
B.S in Industrial Engineering, GPA 3.52/4.0, Student Council President

EXPERIENCE

Graduate Research Assistant, *Human Automation Systems Lab, University of Washington* Sep 2025 – Present
- Designed controlled experiments and user studies to evaluate attention, workload, and performance in complex task environments
- Collected multimodal behavioral and physiological data to understand user states during sustained interaction; generated insights for systems that support sustained attention and reduce cognitive overload

Graduate Research Assistant, *Occupational Ergonomics and Biomechanics Lab, Virginia Tech* Aug 2023 – May 2025
- Led end-to-end mixed-methods research on how knowledge workers interact with computer systems and workplace technologies; conducted interviews, surveys, and behavioral observations alongside sensor-based data collection
- Synthesized qualitative and quantitative data to identify usability barriers and contextual drivers of behavior; analyzed longitudinal data on productivity, comfort, and task engagement

KEY PROJECTS

Driver Experience & Workload in V2X Pedestrian Alert Systems | *City of Bellevue / University of Washington*
Designed and conducted a field study investigating how drivers perceive and respond to connected vehicle pedestrian alerts at crosswalks. Collected physiological measures (E4 Empatica), NASA-TLX workload ratings, and vehicle kinematics; synthesized objective and subjective data to surface usability insights and inform alert design in automated vehicle safety contexts.

Human Vigilance & Cognitive Load Modeling | *University of Washington*
Investigated how cognitive load and alertness influence performance during sustained tasks by integrating behavioral measures with 30+ physiological and contextual features across 10,000+ observations per participant. Developed predictive models improving performance by 20%, generating insights for designing systems that better support attention in safety-critical environments.

NudgeWatch — Adaptive Sit-Stand Nudging System | *University of Washington*
Designed and built the full research instrument myself — a native Apple Watch app with a live posture-sensing engine — to test a context-aware nudging hypothesis no existing tool could support. Ran a 5-participant mixed-methods pilot combining sensor logs, experience-sampling surveys, and retrospective interviews; found the adaptive loop measurably worked but was rarely perceived correctly by users, a key insight for designing trustworthy adaptive systems.

Context-Aware Sit-Stand Intervention for Knowledge Workers | *Virginia Tech*
Conducted a longitudinal field study (3 weeks/participant; 100,000+ records) using interviews, behavioral logs, and sensor data to understand posture-switching behavior and contextual barriers. Applied user clustering to identify behavioral profiles, informing adaptive workplace interventions. Published at HFES 2025; two journal manuscripts in preparation.

SKILLS

Research Methods: Interviews, Surveys, Usability Testing, Longitudinal Field Studies, Experimental Design, ESM
Analysis: Statistical analysis, Thematic coding, Evidence synthesis, Python, SQL
Collaboration: Cross-functional teamwork, Technical communication, Stakeholder presentation

AWARDS

Dean's Fellowship, College of Engineering — University of Washington 2025–2029
Social-Innovation Capability Video Contest, 2nd Place (Data Mining) — Yonsei University 2022
Merit-based Scholarship (Academic Excellence) — Yonsei University 2021